IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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01 April 26

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data collection via API
  - Data collection with Web Scraping
  - Data wrangling
  - Exploratory Data Analysis with SQL
  - Exploratory Data Analysis with Visualization
  - Interactive Visual Analytics with Folium
  - Interactive Dashboard with Plotly Dash
  - Machine Learning Prediction

- Summary of all results
  - Success Rate over Time: An improvement in success rate was observed over time.
  - Success Rate by Launch Site: KSC LC-39A shows the highest success rate of all launch sites.
  - Success Rate by Orbit: ES-L1, SSO, HEO, and GEO were observed to have the highest success rates.
  - Payload: Heavier payloads have a high failure rate early on with a significant improvement over time.
  - Predictive Analysis: The DecisionTreeClassifier algorithm has proven highly accurate in predicting landing outcomes.

Introduction

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars.
- Other providers cost upward of 165 million dollars each.
- Much of the savings is due to SpaceX's ability to reuse the first stage.
- If we can determine if the first stage will land, we can determine the cost of a launch.
- Our company, IBM Developer Skills Network, can use this information to successfully bid against SpaceX for a rocket launch.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology
  - Retrieval and consolidation from multiple SpaceX API endpoints
  - Web scaping tabular data from Wikipedia
- Perform data wrangling
  - Extracted relevant records
  - Flattened fields and resolved missing values
- Perform exploratory data analysis (EDA) using visualization and SQL
  - Visualize variable relationships
  - Look at the data in aggregate
- Perform interactive visual analytics using Folium and Plotly Dash
  - Mark all launch sites on a map
  - Mark successful and failed launches
  - Calculate distances to proximate locations
  - Provide for interactive exploration of the data
- Perform predictive analysis using classification models
  - Build, evaluate, and compare several predictive classification models

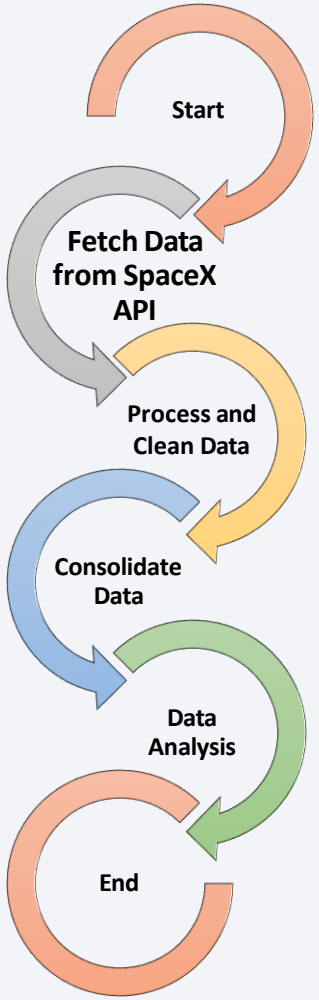
Data Collection

- Data was collected using a combination of retrieval techniques:
  - HTTP requests against various SpaceX API endpoints:
    - Initial launch data was obtained from: **`/v4/launches/past`**
    - Additional data was backfilled from:
      - `/v4/rockets`
      - `/v4/launchpads`
      - `/v4/payloads`
      - `/v4/cores`
  - Tabular data from the List of Falcon 9 and Falcon Heavy launches Wikipedia page.

Data Collection – SpaceX API

- **The initial data was obtained from the `/v4/launches/past` API endpoint.**
- **Additional data was backfilled from the `rocket`, `launchpad`, `payloads`, and `cores` API endpoints, for records with extant corresponding IDs.**

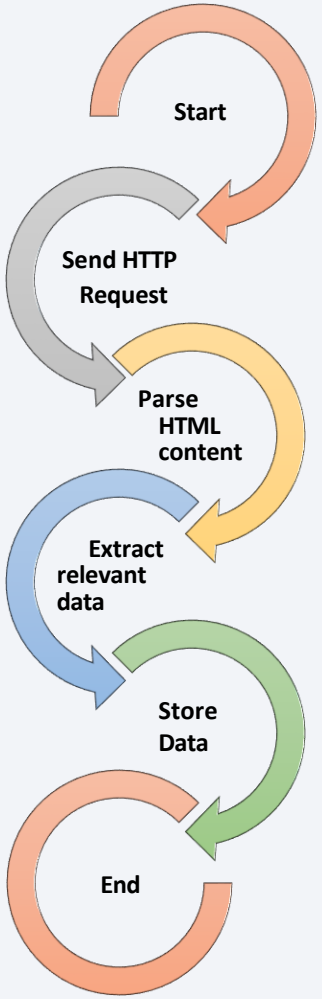
- Notebook (GitHub URL)



Data Collection - Scraping

- Web scraping workflow:
  - Send HTTP Get to Falcon 9 Launch page at Wikipedia, to retrieve the HTML source
  - Create a BeautifulSoup object to extract the tables containing launch data
  - Populate a Pandas DataFrame using the extracted columns

- Notebook (GitHub URL)



Data Wrangling

- Goals:
  - Use Exploratory Data Analysis (EDA) to find patterns in the data.
  - Determine the labels for training supervised models.
- Steps:
  - Find missing values as a percentage of each attribute
  - Identify column types, numerical or categorical
  - Show launches per Launch Site
  - Show distribution of Orbit type in the data set
  - Explore outcomes, and group them by binary outcome (success or failure)
  - Store outcome as "Class" label, to be used as target value in training

- Notebook (GitHub URL)

EDA with Data Visualization

- Visualize relationships to gain insight into the importance of each variable:
  - Flight Number and Outcome
  - Flight Number and Launch Site
  - Payload and Launch Site
  - Orbit and Outcome
  - Flight Number and Orbit
  - Payload and Orbit
  - Yearly success rate
- Expand categorical variables into "dummy" columns
- Convert numerical columns into `float64`

- Notebook ([GitHub URL]()

EDA with SQL

- SQL Queries Performed:
  - Show each unique launch site
  - Show 5 records where launch site names begin with 'CCA'
  - Display the total payload mass carried by boosters launched by 'NASA (CRS)'
  - Display the average payload mass carried by the v1.1 Falcon 9 booster
  - List the date of the first successful ground landing outcome
  - List the booster versions with successful outcomes landing on the drone ship with payloads between 4000kg and 6000kg.
  - List the number of successful and failed mission outcomes
  - List all the booster versions that carried the max payload mass
  - List the month name, outcome, booster version, and launch site for missions with failure outcomes landing on a drone ship in 2015.
  - Show the distribution of outcomes between June 4th, 2010, and March 20th, 2017
- Notebook (GitHub URL)

Build an Interactive Map with Folium

- To find geographical patterns in the data the following items were marked on a map of launch sites:
  - All Launch Sites
  - Successful and Failed Launches
  - Distances between a launch site and proximate landmarks
- Notebook (GitHub URL)

Build a Dashboard with Plotly Dash

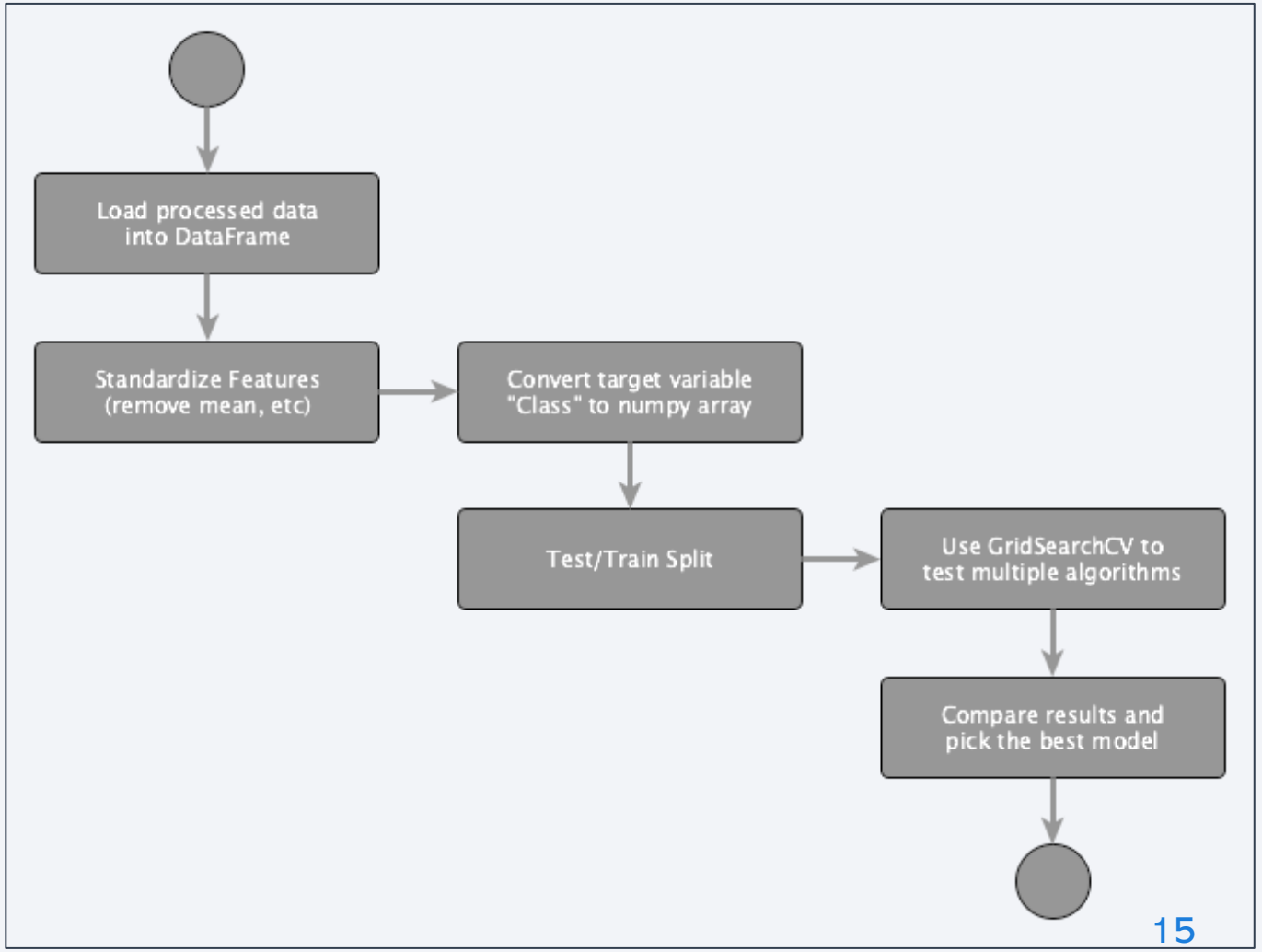
To enable interactive exploration of the data, a Plotly Dash dashboard was developed to include:

- A dropdown selector to choose a Launch Site, affecting:
  - A pie chart
    - All sites selected: shows the breakdown of successful outcomes across all sites
    - A launch site selected: shows the breakdown of successful vs failed launches for the given site
  - A scatter plot
    - All sites selected: shows outcome by payload mass and booster version for all sites
    - A launch site selected: shows outcome by payload mass and booster version for the given site
- A Payload Mass range selector that filters data points on the scatter plot

- Plotly Dash App (GitHub URL)

Predictive Analysis (Classification)

- Load data
- Apply StandardizedScaler on X
- Convert Y to numpy array
- Split training and testing data
- Use GridSearchCV to test hyperparameters for multiple algorithms:
  - Logistic Regression
  - SVC
  - Decision Tree Classifier
  - K Neighbors Classifier
- Notebook (GitHub URL)



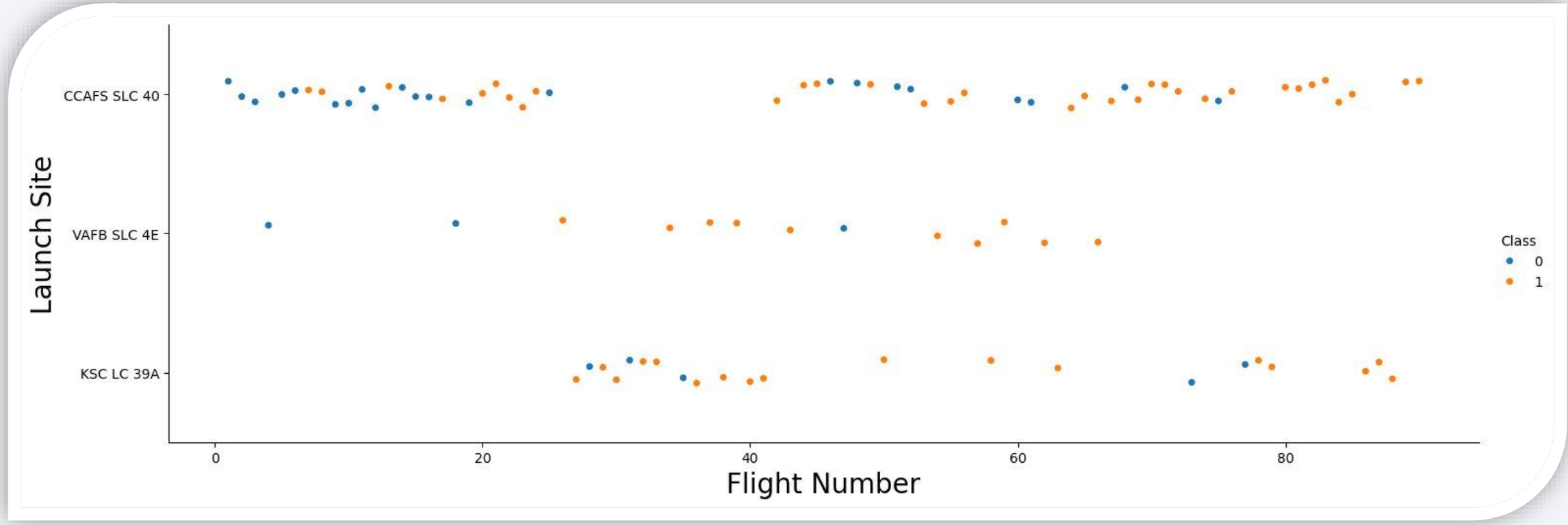
Results

- **Exploratory Data Analysis (EDA) Results**
  - **Mission Success Over Time**
    - Success rate improved significantly over time, with early launches having higher failure rates and more recent
    - launches showing consistent success.
  - **Launch Site Performance**
    - **KSC LC-39A** had the highest success rate, showing strong performance in Falcon 9 booster landings.
  - **Orbit Type Success**
    - Some orbits, like **GEO** and **SSO**, showed a perfect success rate, suggesting certain mission types are more
    - reliable.
- **Interactive Analytics Demo**
  - **Launch Site Map**
    - Interactive map showing the geographic locations of launch sites.
  - **Mission Success Dashboard**
    - A dashboard visualizing success rates across launch sites and orbit types.
- **Predictive Analysis Results**
  - **Best Model**
    - **Decision Tree** model showed the best performance with **94.44% accuracy** in predicting launch success.
  - **Key Metrics:**
    - Low number of false positives/negatives, making the model reliable for future predictions.

Section 2

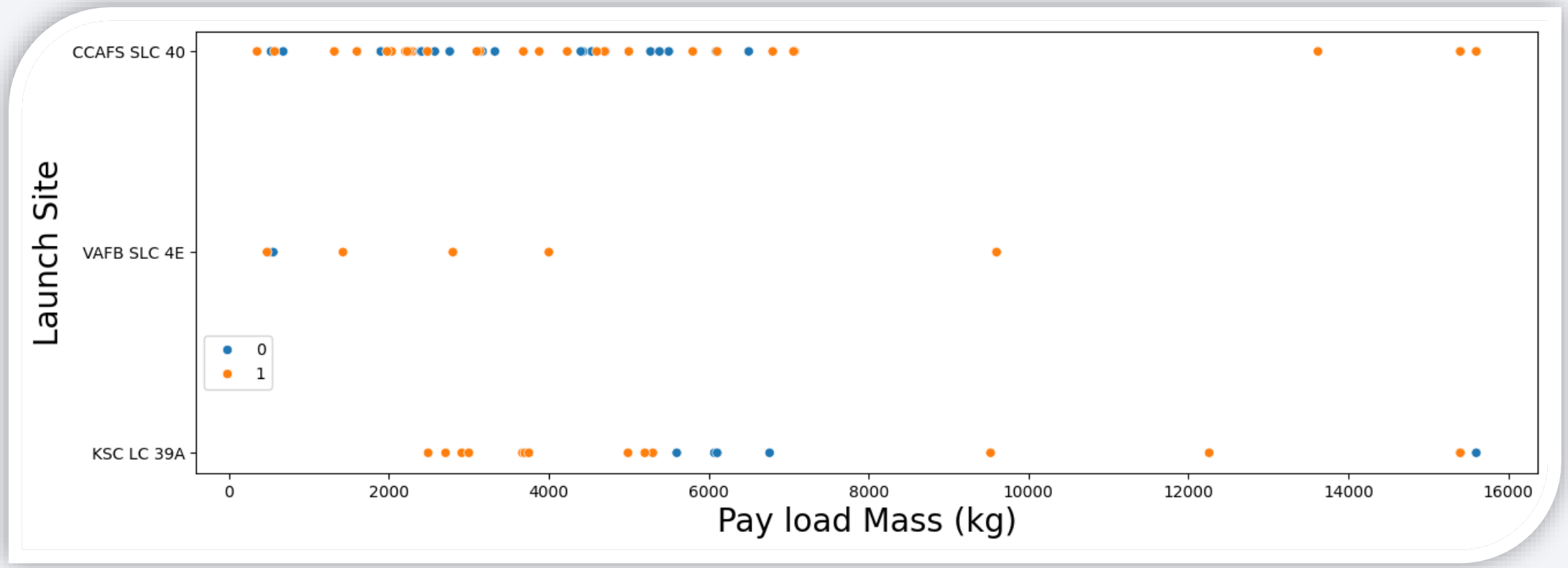
Insights drawn from EDA

Flight Number vs. Launch Site



- All sites show a mix of first stage landing successes and failures, with successes increasing over time.
- Early flights predominantly resulted in failures, indicating improvements to technology or process.
- While CCAFS SLC 40 has the most total flights, VAFB SLC 4E appears to have a relatively higher proportion of successful landing outcomes.

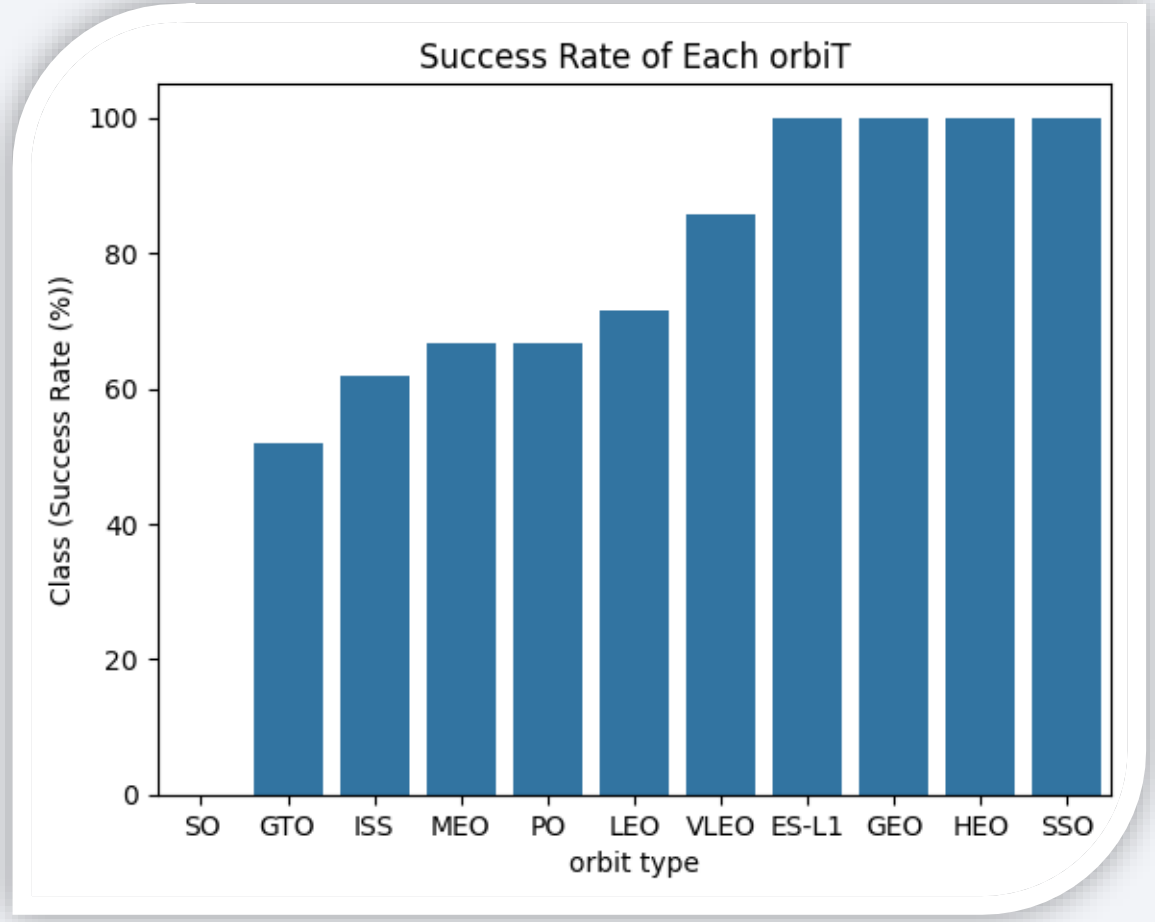
Payload vs. Launch Site



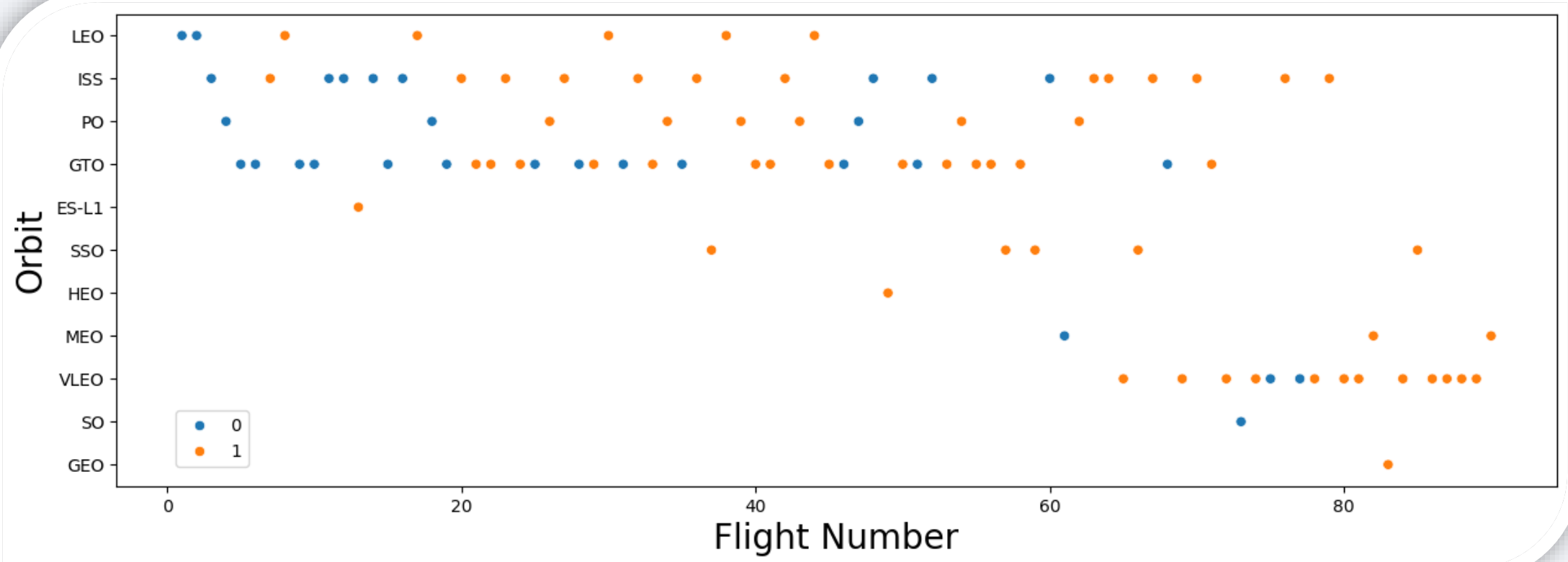
- All sites show a wide range of payload weights, from light to heavy payloads.
- Early flights trend toward lighter payloads, representing the bulk of the landing failures.
- This suggests technological or operational improvements lead to a greater rate of success with heavier payloads.

Success Rate vs. Orbit Type

- Some orbits, such as ES-L1, SSO, HEO, and GEO consistently show high success rates.
- Others such as GTO show more mixed outcomes, suggesting some orbit types may introduce operational or technological challenges.
- With only one launch, there is not enough data for the SO orbit type to provide an accurate analysis.

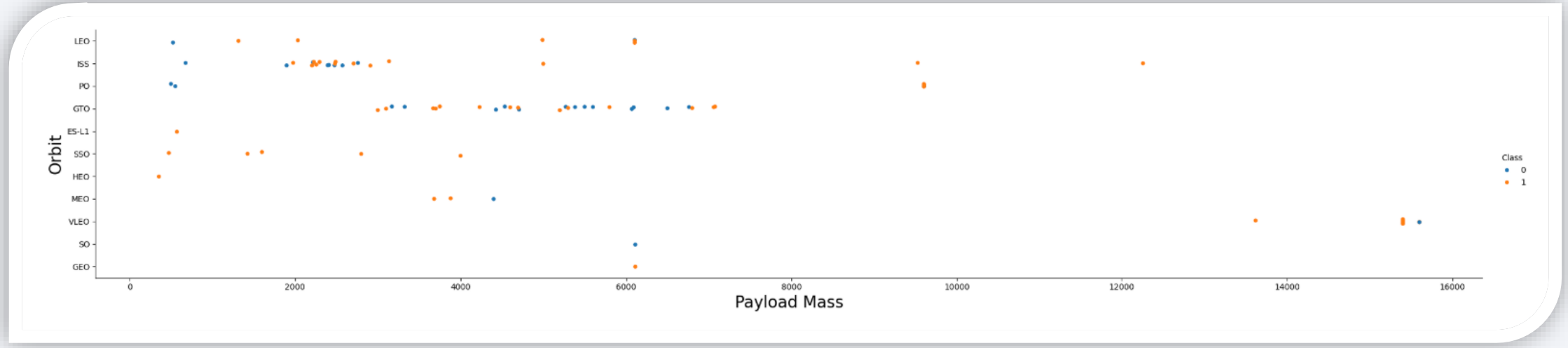


Flight Number vs. Orbit Type



- Many orbits are represented throughout the flight number range, some orbits are not attempted until later flights.
- There is a noticeable improvement in landing success as flight numbers increase, indicating an accumulation of experience and ongoing improvements.

Payload vs. Orbit Type



Explanation:

- **X-axis (Payload Mass):** The payload mass refers to the weight of the cargo that the rocket transports into orbit.
- **Y-axis (Orbit):** Classification of the different orbits.

Insights:

- LEO (Low Earth Orbit) supports a wide range of payload masses.
- GEO (Geostationary Orbit) missions typically carry heavier payloads.
- SSO (Sun-Synchronous Orbit) and other specialized orbits handle lighter payloads.
- Payload mass impacts mission complexity.

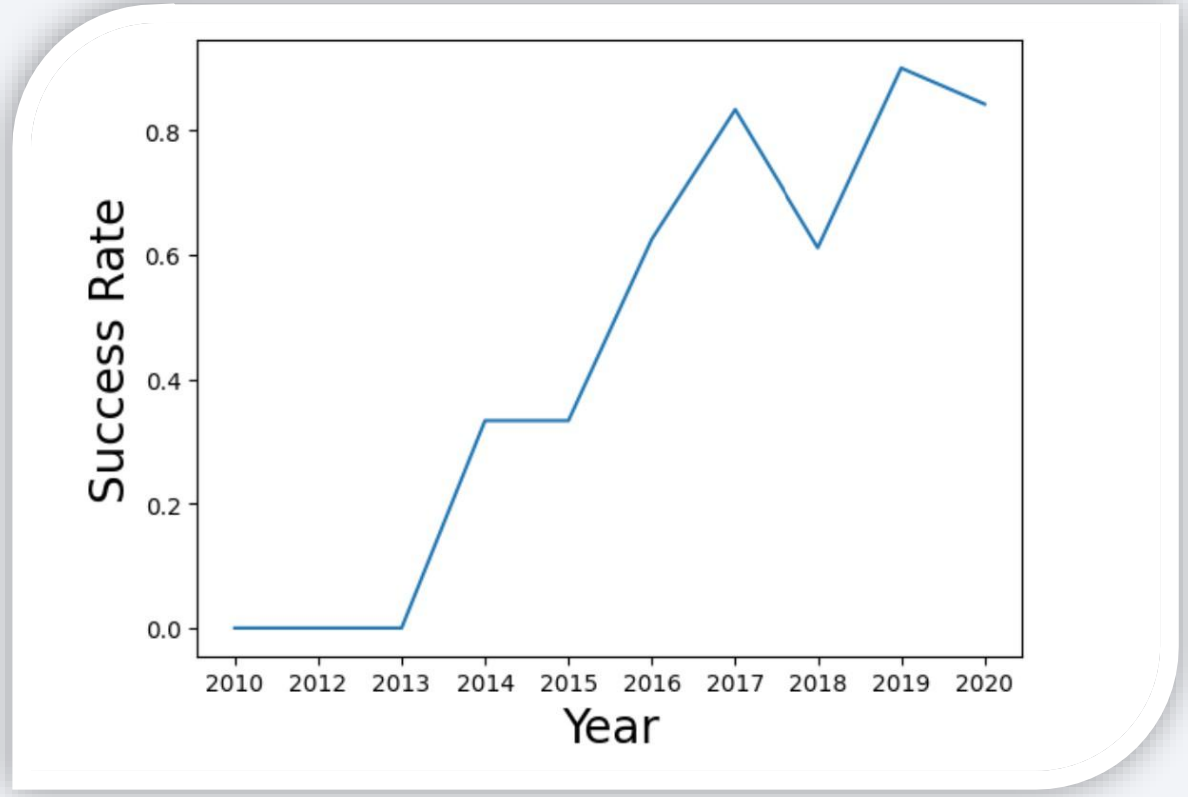
Launch Success Yearly Trend

Explanation:

- **X-axis (Year):** Year of different launch.
- **Y-axis (Success Rate):** Percentage of success of launch.

Insights:

- The most success rate is 2016 and 2019.
- Since 2019 have gone down the success rate.



All Launch Site Names

There are four unique Launch Sites

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

```
%sql SELECT DISTINCT launch_site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

```
SELECT DISTINCT Launch_Site from SPACEXTABLE;
```

Launch Site Names Begin with 'CCA'

First five records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Total Payload Mass

The total payload carried by boosters from NASA (CRS) is 45,596kg.

```
SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD
  FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
%sql SELECT SUM(payload_mass__kg_) AS total_payload_mass FROM SPACEXTABLE WHERE customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

total_payload_mass
45596

Average Payload Mass by F9 v1.1

```
1]: %sql SELECT AVG(payload_mass__kg_) AS average_payload_mass FROM SPACEXTABLE WHERE booster_version like '%F9 v1.1%';

 * sqlite:///my_data1.db
Done.
```

1]:

average_payload_mass
2534.6666666666665

The average payload mass carried by booster version F9 v1.1 is 2,534.67kg.

```
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD_MASS
  FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

First Successful Ground Landing Date

```
30]: %sql SELECT min(date) AS first_successful FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

 * sqlite:///my_data1.db
Done.
```

30]:

first_successful
2015-12-22

The first successful landing outcome on ground pad occurred on December 22nd, 2015.

```
SELECT MIN(Date) as LaunchDate
  FROM SPACEXTABLE
  WHERE Landing_Outcome = 'Success (ground pad)';
```

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS__KG_ BETWEEN 46
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:

```
SELECT Booster_Version, PAYLOAD_MASS__KG_
  FROM SPACEXTABLE
  WHERE Landing_Outcome = 'Success (drone ship)'
    AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Booster	Payload Mass
F9 FT B1022	4,696kg
F9 FT B1026	4,600kg
F9 FT B1021.2	5,300kg
F9 FT B1031.2	5,200kg

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Total number of successful and failure mission outcomes

Mission Status	Count
Failure	1
Success	100

Boosters Carried Maximum Payload

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

The maximum payload sent was 15,600kg.

The boosters that carried the maximum payload are:

```
SELECT
  DISTINCT Booster_Version,
  PAYLOAD_MASS__KG_
FROM SPACEXTABLE
  WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
  )
ORDER BY Booster_Version;
```

Booster Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

2015 Launch Records

```
%%sql SELECT strftime('%m', Date) AS month, Date, booster_version, launch_site, landing_outcome
    FROM SPACEXTABLE
    WHERE landing_outcome = 'Failure (drone ship)' AND strftime('%Y', Date) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

List of failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015:

```
SELECT
   CASE strftime('%m', Date)
      WHEN '01' THEN 'January'
      WHEN '02' THEN 'February'
      WHEN '03' THEN 'March'
      WHEN '04' THEN 'April'
      WHEN '05' THEN 'May'
      WHEN '06' THEN 'June'
      WHEN '07' THEN 'July'
      WHEN '08' THEN 'August'
      WHEN '09' THEN 'September'
      WHEN '10' THEN 'October'
      WHEN '11' THEN 'November'
      WHEN '12' THEN 'December'
   END as Month,
   Landing_Outcome, Booster_Version, Launch_Site, Date
FROM SPACEXTABLE
WHERE strftime('%Y', Date) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Month	Outcome	Booster	Launch Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT Landing_Outcome, COUNT(*) AS COUNT_LAUNCHES FROM SPACEXTABLE WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' G
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	COUNT_LAUNCHES
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

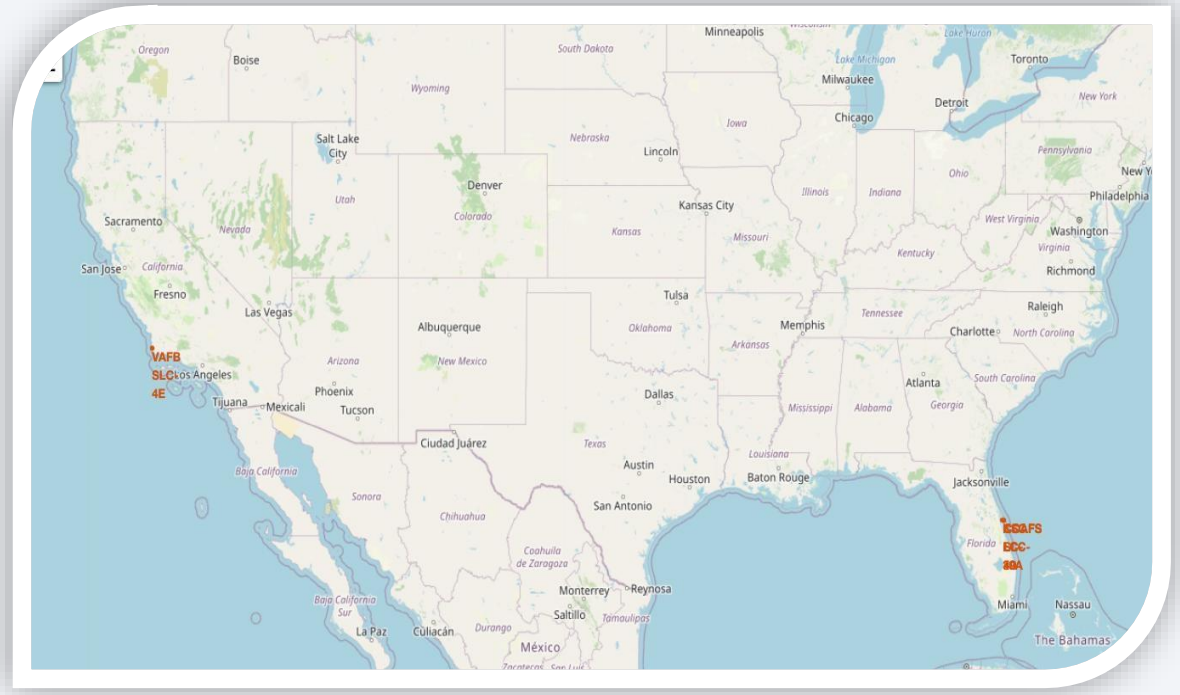
Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

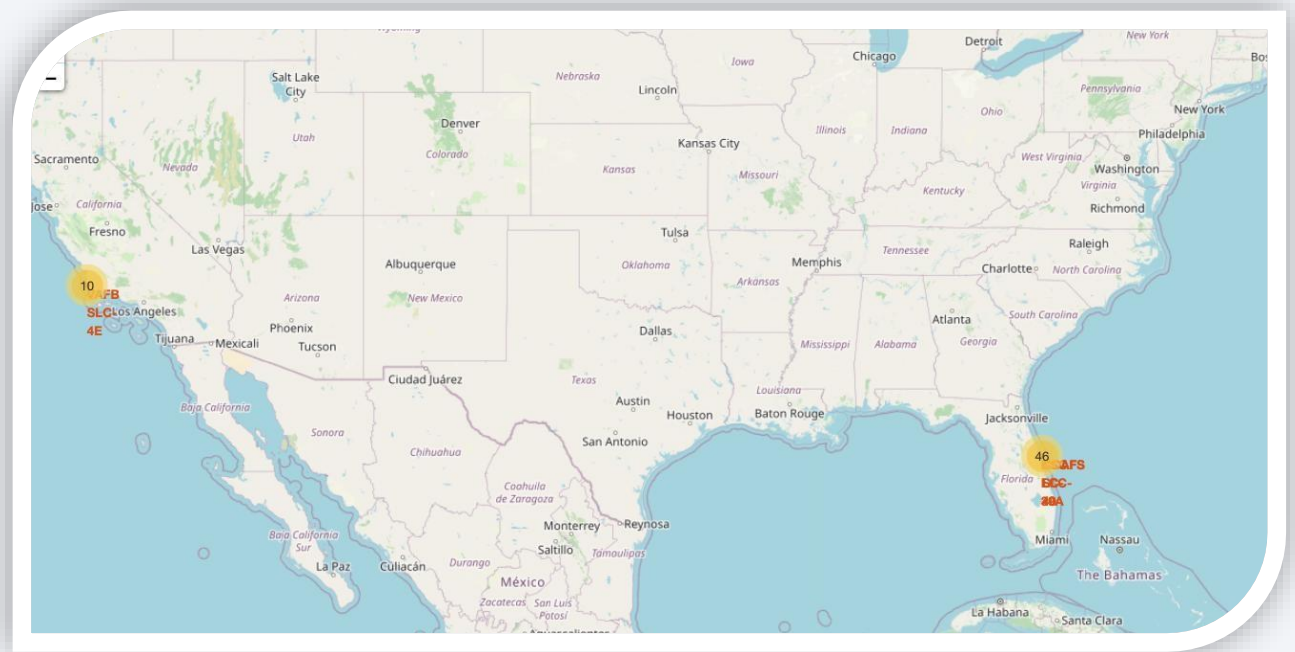
Locations of all launches sites in the USA

- As expected, most launchers in the United States are located in Florida because this point is closer to the equator, which allows them to be more in sync with the Earth's rotation, make safer launches and save on costs.
- Also for safety reasons, launchers in the United States are placed in areas facing the ocean so that the launch is in that trajectory and not towards populations. In other countries, this safety measure is taken in desert or uninhabited areas.



Total releases in each location in the USA

- Because there are more launchers and availability in Florida due to cost, maintenance and mission issues, most are launched on rocket launchers in that area.



Launch records in the main launcher site

- Brands use the following color
  - Green are successful launches
  - Red are unsuccessful launches
- Launch Site KSC LC-39A has a very high Success Rate.

Distance from the launch side KS LC-39A

- **Proximity to Key Infrastructure:**
  - **Railway:** 15.23 km
  - **Highway:** 20.28 km
  - **Coastline:** 14.99 km
  - **Closest City (Titusville):** 16.32 km
- **Risk Consideration:**
  A failed rocket at high speed can travel **15-20 km in seconds**, posing potential risks to populated areas.
- **Insight:**
  Location balance between **accessibility & safety** is crucial for optimizing future launch operations.

Distance from the launch side KS LC-39A

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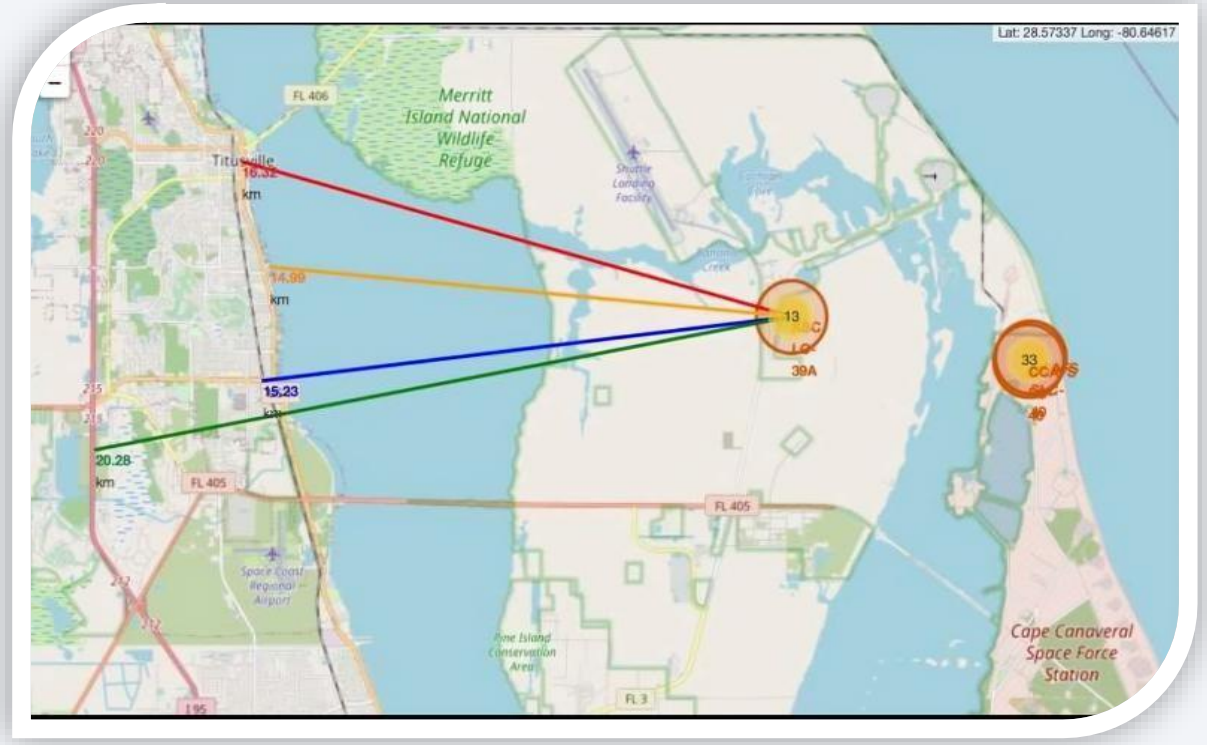
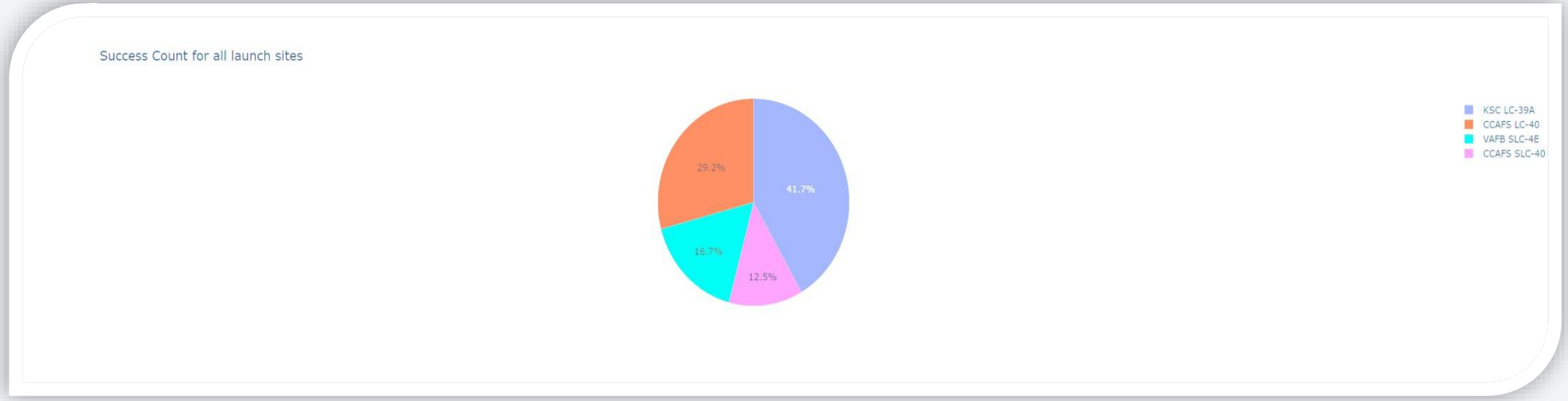


Section 4

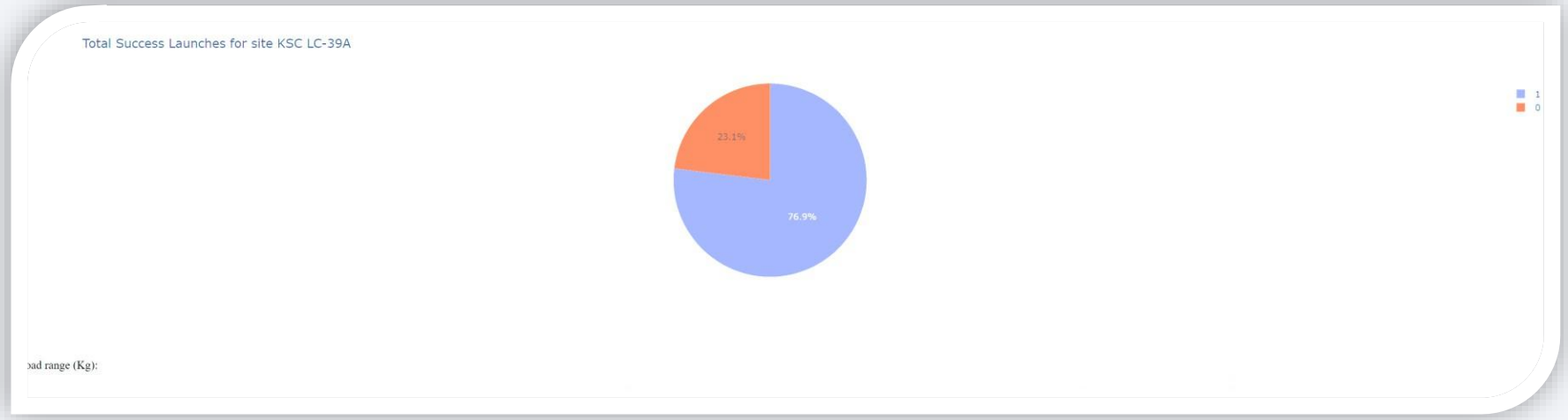
Build a Dashboard with Plotly Dash

Success launch percentage for all sites



- This pie chart shows the success rates for rocket launches at each of the launch sites in the United States.
- It can be seen that KSC LC-39A has the highest number of successful launches.
- CCAFS SLC-40 has the lowest number of successful launches.
- Nothing can be concluded from this, except that the most common areas for successful launches.

Success launch percentage in KSC LC-39A



- You can see in this pie chart the percentage of failure and success rates in this location.
- Obviously if you look at the rest, this is the one with the highest success rate compared to the rest of the cases.
- Also, its failure rate is much lower than the average, so investigating why fewer failures occur here would be essential.

PAYLOAD MASS VS. LAUNCH OUTCOME FOR ALL SITES

- Here is the list of all the launches in any area relating the payload mass and launch outcome divided into two sections: the first graph is the section from 0 to 5000 and the second section, from 5000 to 10000 kg.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- **Test Set scores are identical across models** (Logistic Regression, SVM, Decision Tree, KNN), making it unclear which performs best.

- **This may be due to the small test sample size (18 samples).**

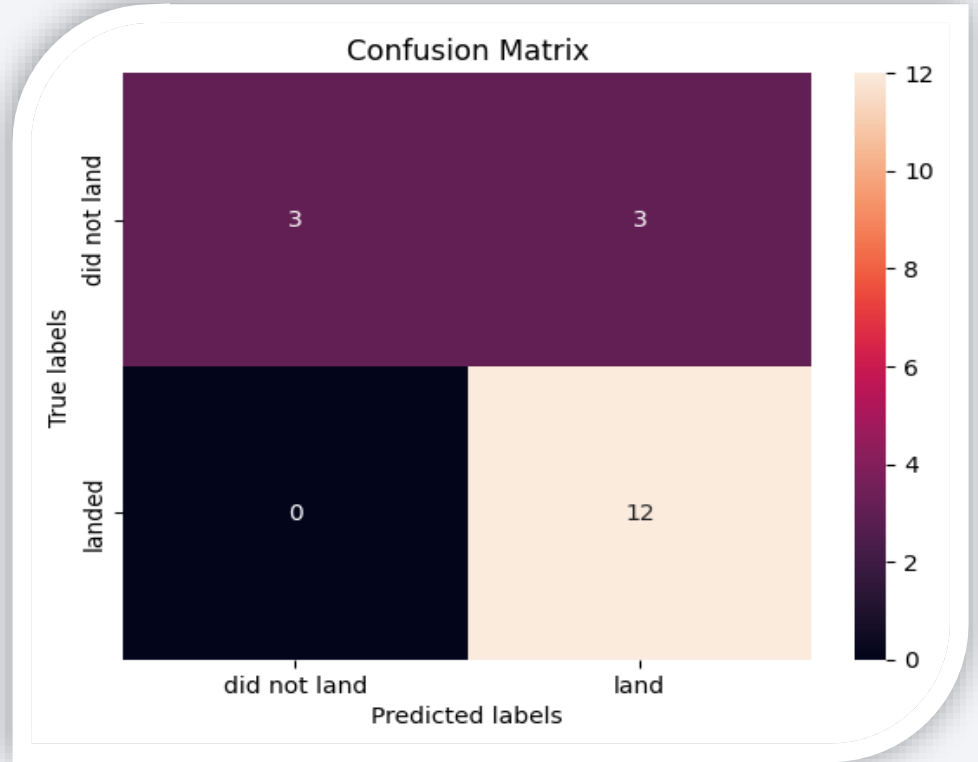
- **To address this, we tested all models on the full dataset.**

- **Results confirmed Decision Tree as the best model**, achieving the highest accuracy and performance.

	Model	Jaccard Score	F1 Score	Accuracy Score
0	Logistic Regression	0.8	0.888889	0.833333
1	Support Vector Machine	0.8	0.888889	0.833333
2	Decision Tree	0.8	0.888889	0.833333
3	K-Nearest Neighbors	0.8	0.888889	0.833333

Confusion Matrix

- **Logistic Regression effectively distinguishes between classes**, showing its predictive capability.
- **True Positives & True Negatives are well-classified**, indicating good model performance.
- **Some misclassifications exist**, but further tuning can improve accuracy.
- **Overall, the model provides reliable predictions for launch success.**



Conclusions

- **Decision Tree Model performed best** in predicting launch success.
- **Lighter payloads have higher success rates**, while heavier payloads face more challenges.
- **Launch sites are strategically located** near the **Equator and coastlines** for efficiency and safety.
- **Launch success rates have improved over time**, reflecting advancements in technology and operations.
- **KSC LC-39A has the highest success rate**, making it the most reliable launch site.
- **Certain orbits (ESL-1, GEO, HEO, and SSO) achieved 100% success**, indicating mission reliability in these destinations.
- **Future Improvements**
  - **AI-driven predictive maintenance:**
    Using **machine learning to detect potential booster failures** before launch, reducing the risk of malfunctions.
  - **Optimizing payload-to-orbit matching:**
    Enhancing **payload distribution strategies** to ensure each rocket is optimized for its target orbit, improving efficiency and success rates.
  - **Key Takeaway:**
    With advanced AI and smarter launch planning, **SpaceX can further increase reliability, reduce failures, and improve mission success.**

Appendix

1. **Data Collection:**
    1. **Public APIs:** Used SpaceX public APIs to retrieve detailed launch data, including rocket specifications, payload information, and landing outcomes.
    2. **Web Scraping:** Implemented web scraping techniques on sources like Wikipedia to complement and validate API data.
2. **Data Preparation & Cleaning:**
    1. **Handling Missing Values:** Processed and cleaned the dataset to ensure consistency and completeness.
    2. **Feature Engineering:** Created new variables to improve predictive modeling accuracy.
3. **Exploratory Data Analysis (EDA):**
    1. **Data Visualization:** Used **matplotlib, seaborn, and Plotly** to analyze trends in launch success rates, payload mass, and orbit types.
    2. **SQL Queries:** Performed **SQL-based analysis** to extract insights from structured launch data.
4. **Predictive Modeling:**
    1. **Machine Learning Models:**
        1. **Decision Tree & Logistic Regression** were tested for predicting successful landings.
        2. **Hyperparameter Tuning** improved model accuracy.
    2. **Evaluation Metrics:** Accuracy, precision, recall, and F1-score were used to assess model performance.
5. **Interactive Analytics & Visualization:**
    1. **Folium Maps:** Used for geographic visualization of launch sites and risk analysis.
    2. **Plotly Dash:** Created an interactive dashboard to explore SpaceX launch data dynamically.

Thank you!